Center for Advanced Logistics and Focused Analytics

Presents

The Urban Autonomous Mobility Challenge (UAMC)

Introduction

Imagine a city street shrunk down to a tabletop scale of lanes, intersections, traffic lights, pedestrian crossings, and parking bays and a vehicle you built and programmed yourself finding its own way through it, with no remote control and no human hand guiding it.

The Urban Autonomous Mobility Challenge (UAMC) invites student teams to design, build, and program a scaled autonomous vehicle capable of navigating exactly this kind of miniature urban environment, entirely on its own. Inspired by the Bosch Future Mobility Challenge, UAMC is redesigned to run fully within Techfest, IIT Bombay's 3-day festival window, with a remote qualification stage held 6–8 weeks in advance to ensure every team arrives with a working, tested vehicle rather than a work-in-progress.

The challenge rewards genuine autonomous perception and decision-making lane-keeping, sign and signal recognition, obstacle handling, and parking built using accessible hardware and sound engineering practice. Teams that qualify get the exact on-site arena layout in advance, so the festival days are spent proving readiness, not discovering problems for the first time.

Core theme: "A self-driving car navigating a smart city street."

General Elements Expected Across the Build

To keep the challenge accessible, safe, and genuinely autonomous, every vehicle is expected to incorporate the following, with higher evaluation weight given to teams that integrate these well:

- **Hardware:** Any single-board computer or microcontroller (Raspberry Pi, Jetson Nano, ESP32, Arduino, etc.) paired with cameras, ultrasonic, IR, low-cost LiDAR, IMU, or wheel-encoder sensors.

- **Autonomy ("Physical AI"):** Genuine onboard perception and decision-making lane detection, sign/signal recognition, obstacle avoidance, or adaptive parking with zero live remote control during a timed run.
- **Engineering Documentation:** A clear account of system architecture, perception approach, and control stack, submitted as a Technical Design Document (TDD) at the qualification stage.
- **Safety:** A demonstrable, working emergency stop; fused/current-limited battery output; no exposed sharp edges or unshielded high-torque parts.
- **Scalability & Usability:** A cost-effective, well-engineered build that could plausibly generalize beyond the competition arena.

Theme: Building for the Smart City Street

Every run takes place in a single, unified urban arena that combines the full range of real-world driving scenarios: lane discipline, intersections, pedestrian safety, parking, and (in later rounds) dynamic traffic. Rather than choosing between separate themes, teams build one vehicle capable of handling the complete urban circuit as it escalates in complexity round by round.

Problem Statement

Design and build a scaled autonomous vehicle that can navigate a miniature urban street network following lanes, obeying stop signs and traffic lights, stopping for pedestrians, parking within marked bays, and avoiding obstacles without any human intervention during a timed run. Using accessible hardware (e.g., Raspberry Pi, Arduino, Jetson Nano, camera and ultrasonic/IR sensors), build a system that autonomously perceives the road environment and reacts to it correctly and safely. Where relevant, back your design with basic engineering calculations (e.g., turning radius, sensor thresholds, stopping distance) to show your solution is not just working by trial and error but is understood and tunable.

The following are provided purely for inspiration; teams are encouraged to go beyond them with their own approaches to perception, control, or mechanical design:

- A vehicle using a single camera and classical computer vision (edge/colour detection) for lane-keeping and stop-sign recognition.
- A vehicle combining a camera with ultrasonic sensors for simultaneous lane-following and static-obstacle avoidance.

- A vehicle using a lightweight neural network (e.g., a small CNN) for end-to-end steering, trained on data collected from the practice track.
- A vehicle with a dedicated parking routine that uses IR/ultrasonic ranging to detect an open bay and align itself within the boundary lines.

Resources (For Inspiration Only)

1. Lane Detection using Open CV and Raspberry Pi:https://www.youtube.com/watch?v=example1
2. Self-Driving RC Car with Arduino — Lane Keeping: https://www.youtube.com/watch?v=example2
3. Traffic Sign Recognition with a Pi Camera: https://www.youtube.com/watch?v=example3
4. Autonomous Parking Robot using Ultrasonic Sensors: https://www.youtube.com/watch?v=example4
5. Jetson Nano JetBot — Basic Autonomous Navigation: https://www.youtube.com/watch?v=example5

YouTube Channels for Inspiration

1. Circuit Digest: https://www.youtube.com/@CircuitDigest
2. How To Mechatronics: https://www.youtube.com/@HowToMechatronics
3. DroneBot Workshop: https://www.youtube.com/@DroneBotWorkshop
4. JetsonHacks: https://www.youtube.com/@JetsonHacks
5. How To Electronics: https://www.youtube.com/@HowToElectronics

Evaluation Criteria

- **Autonomy & Perception (30%):** Genuine onboard decision-making — lane-keeping accuracy, sign/signal recognition, and obstacle response — without any live remote control.
- **Reliability & Robustness (25%):** Consistent, repeatable performance across the arena's lanes, intersections, crossings, and parking bay.
- **Engineering Soundness (20%):** Sound calculations and reasoning behind sensor thresholds, control logic, and mechanical design, clearly documented in the TDD.
- **Safety Compliance (15%):** A working emergency stop, safe wiring, and adherence to all vehicle safety specifications.
- **Scalability & Usability (10%):** Cost-effectiveness, build quality, and potential for the approach to generalize beyond the competition arena.

Structure

The competition consists of a remote qualification stage followed by on-site rounds at Techfest, structured to fit entirely within the 3-day festival window:

Round 1: Qualification — Video Submission & Technical Design Document (Online)

Submit a video (3–5 minutes) of your vehicle completing a basic lane-following and stop-at-obstacle task on a self-built practice track, along with a Technical Design Document (TDD, 1–2 pages). Based on this, Top 25 teams are invited on-site.

Requirements:

- Practice-track video showing autonomous lane-keeping and a controlled stop at an obstacle, with no visible remote control.
- System architecture: sensors, compute, and control stack used.
- Perception approach: how lane, sign, or signal detection is implemented.
- Safety design: E-stop implementation and failure handling.

Submission: All online submissions for this competition must be made through the official Techfest website prior to the specified deadline. Only the Team Leader is authorized to submit on behalf of the team. Submissions made by any other team member will not be considered valid, multiple submissions from the same team will not be accepted and may lead to disqualification from the competition.

Round 2: On-Site Elimination Heats (Techfest, Day 1–2)

Qualified teams arrive on Day 1 for check-in, technical inspection, and a fixed practice slot on the actual arena, followed by heats on Day 2:

- Basic Navigation Heat: Lane-keeping, one stop-sign intersection, one pedestrian crossing, one static obstacle. The top 60% of teams advance.
- Intermediate Navigation Heat: Adds a traffic-light-controlled intersection, a roundabout, and autonomous parking. Top 8–10 teams advance to Round 3.

Round 3: Semifinal & Grand Finale (Techfest, Day 3)

- Semifinal: Full urban circuit combining every element from earlier rounds, run back-to-back and ranked by score; top 4–6 teams advance.
- Grand Finale: Full circuit plus a dynamic obstacle/overtaking segment and a randomized surprise rule (e.g., a mid-run detour) announced only at the start of the day, to test genuine generalization rather than a memorized path

Winners are determined by total score across positive points, penalties, and time-based tiebreakers, with results announced the same evening.

Team Specifications and Eligibility

- **Eligibility:** Open to undergraduate/postgraduate students with a valid ID from any recognized institution. Teams may include students from different institutes.
- **Team Size:** 3–5 members per team, with one designated Team Lead, fielding only one vehicle.
- **Mentors:** Faculty or industry mentors may guide teams but must not be present at, or assist during, any timed run.

Competition Timeline

Event	Important Dates
Registration Deadline	October 31st
Round1 (Qualification) Submission	November 10th
Round 1 Results	November 17th (24)
Final Arena Layout Released	November 24th (31)
Round 2 (Elimination Heats)	16-17th December
Round 3 (Semifinal & Grand Finale)	18th December

Scoring System

Scoring is point-based: positive points for correctly completed maneuvers, penalties for violations, and a time-based tiebreaker.

Positive Points

Task	Points
Successful lane-keeping (per road segment)	10
Correct stop-sign behaviour (full stop + proceed on right-of-way)	15
Correct traffic-light behaviour (stop on red, proceed on green)	20
Correct pedestrian-crossing stop	20
Correct roundabout entry/exit	15
Successful obstacle avoidance (static)	15
Correct speed-zone compliance	10
Successful overtake (Finals only, safe & legal)	25
Full circuit completion without human intervention	+30 bonus

Penalties

Violation	Penalty
Lane departure (wheels fully outside lane)	5 per occurrence
Missed/ran a stop sign	5
Ran a red light	10
Failure to stop for pedestrian	10
Collision with static obstacle/prop	10
Collision with dynamic obstacle/other vehicle	15
Illegal/unsafe overtake	20
Parking outside bay lines	5 (partial credit if partially inside)
Manual human intervention (touching vehicle)	5 per touch; run may be terminated at 3rd touch
Exceeding time limit	Scored on progress up to time-out; no completion bonus
Spec violation found post-run	Disqualification from that round

Tiebreakers

1. Higher total points.
2. Fewer human interventions.
3. Faster completion time.
4. Sudden-death shootout run (Finals only, if still tied).

General Rules

1. Teams must register online, receiving a Team ID for future reference.
2. Registration and Round 1 (qualification) submissions close on the published deadline; no late entries are accepted.
3. The organizers' and judges' decisions are final and binding. Techfest reserves the right to verify participant and vehicle details.
4. Techfest, IIT Bombay, is not responsible for late, lost, or corrupted entries.
5. Vehicle designs and software must be original and not infringe on existing patents, copyrights, or publications.
6. No live remote control or teleoperation is permitted during any timed run; violation results in disqualification from that round.
7. A working, demonstrable emergency stop is mandatory; a non-functional E-stop is grounds for exclusion from the run.
8. No rule changes will be made once Day 1 technical inspection begins. Clarifications issued before that point are posted to the official communication channel and are binding.
9. The latest information will be available on the official Techfest website, with registered participants notified of changes via email.
10. Official communication for important announcements will be through the satwik@techfest.org. Queries submitted via email will not be entertained.

Certificate Policy

1. The top three teams in the Grand Finale receive a Certificate of Excellence.
2. Only teams that are shortlisted for on-site rounds and present their vehicle at Techfest are awarded an e-Certificate of Participation.

Prize Money

Prize money will be awarded to the Winner, Runner-up, and Second Runner-up via NEFT, following the results of the Grand Finale. Winners must email the following to the official UAMC contact immediately after results are announced:

Subject: "UAMC, [Team ID] – [Position]" (e.g., "UAMC, UM-211234 – 1st Position")

Body:

1. Account Holder's Name
2. Account Number
3. Bank Name and Branch
4. IFSC Code
5. A photograph of the Bank Passbook as proof